

# Design Choices for the Tripartite Proton Junction

$$W_{ABC}$$

distinct holes, the relaxation fix, and why the singlet needs complex geometry

Tessera — cobordism subsystem (#396)

## Abstract

The companion note showed that a *bipartite* merge cannot produce the color-singlet proton: the singlet is the three-index alternating invariant  $\varepsilon_{ijk}$ , absent from  $\mathbf{3} \otimes \mathbf{3}$ . We therefore build the proton on a *trivalent* junction  $W_{ABC}$  with boundary  $\partial W = \psi_A \sqcup \psi_B \sqcup \psi_C$ . This note records the four design choices made in that construction and shows that each is forced by the same fact — the singlet is irreducibly a *three-body* object. (1) The three inputs must enter as *distinct* register holes (independent 1-cycles), never as stacked layers of one shared register, or the relaxation averages them. (2) The metric relaxation must *gate* the state-residual gradient once the inputs are carried, or an ill-conditioned step explodes. (3) A van Raamsdonk metric built from *pairwise* mutual information cannot select the singlet, because the singlet’s pairwise MI is maximal and *uniform* — its content is a three-party correlation. (4) The  $\mathbb{Z}_3$  phases of  $[1, \omega, \omega^2]$  live in the complex boundary state; to let the *geometry* carry them the metric must go *complex*, i.e. Lorentzian — which simultaneously realizes the photon channel (null worldline edges).

## 1 The junction

The register is the homology of a holed surface. For the proton we take one connected base surface — a 2-frequency geodesic icosahedron ( $S^2$ , 42 vertices) — minus 12 vertex-disjoint hole triangles grouped into **four windows of three**:  $A, B, C$  (the three color inputs) and  $R$  (the emergent result). Extruding  $\times I$  (**prismCells**) gives a connected 3-complex  $W$  with  $b_1 = 11$ , a genuine valid manifold (every triangle in  $\leq 2$  tetrahedra, **dualComplexValid**) — not a weld. Each window’s three color holes carry a color amplitude triple; the carried object is a color rep, and charge is the oriented sum  $\Sigma(\psi) = \sum_i \psi_i$  ( $\Sigma = 0$  is color-neutral).

## 2 Choice 1 — distinct holes, not stacked layers

A single *shared* register (the bipartite construction, and the linear multi-layer staircase) fails for a precise reason.

**Principle 1** (Shared registers average). *In a product complex  $M \times I$  the copy of a given hole-circle on each layer is the same homology class (the layers cobound the tube wall). Pinning three inputs on one shared class asks a single period to equal three distinct targets; **residualForPeriods** returns the least-squares solution — the average — which retains the symmetric (sextet) component and is never the totally antisymmetric singlet.*

This is why the “ $b_1$  question” is a false binary: both the  $b_1 = 2$  linear staircase ( $S^2 - 3$  holes)  $\times [0, 3]$  and the  $b_1 = 3$  four-holed sphere stack the inputs on one shared cycle, and both average. What

carries the singlet is the opposite: the three inputs as *independent* 1-cycles — distinct holes on the base surface. Confinement is then *conservation at the junction*: on a connected surface-minus-holes the global Stokes relation forces

$$\sum_{\text{all holes}} (\text{induced period}) = 0 \quad \implies \quad \Sigma_R = -\Sigma_{\text{inputs}}.$$

Three color-neutral inputs ( $\Sigma = 0$  each) therefore force a color-neutral result. Measured on the built junction: the three neutral pairs carry *exactly* ( $r_U \sim 10^{-27}$ , independent cycles — no over-determination); the emergent result is neutral for neutral inputs ( $|\Sigma_R| \sim 0$ ) and carries the net charge for colored inputs ( $|\Sigma_R| \approx 3$ ); and the construction is  $S_3$  (color-relabel) invariant.

### 3 Choice 2 — gate the state-residual gradient

The junction relaxation initially stalled at iteration 0 (no improving step) where the smaller bipartite relaxed fine. The cause is a direct consequence of Choice 1.

The relaxation minimizes  $\beta \|\nabla_I S\|^2 + r_{\text{state}}$  over the interior edge lengths by a Gauss–Newton / Levenberg–Marquardt step. Because the inputs are carried *exactly* on independent cycles,  $r_{\text{state}} \sim 10^{-27}$  on *any* metric — so its analytic gradient is numerical noise. Folded into the step through the (ill-conditioned) action Gauss–Newton Hessian, that noise gradient explodes the step,  $\|\delta\| \sim 10^{13}$ , and every line-search trial diverges.

**Remark 1** (The fix). Fold the state-residual gradient into the step *only* while  $r_{\text{state}}$  is above its convergence floor; once the state term is converged, minimize the action alone (and guard the line search against a degenerate-step throw). The bipartite/operator paths ( $r_{\text{state}} \gg \varepsilon$ ) are unchanged; the junction now relaxes ( $0 \rightarrow 40^+$  iterations,  $\|\nabla S\|^2 : 315 \rightarrow 14$ ). The action-only step is provably a descent direction — the explosion was entirely the spurious state gradient.

### 4 Choice 3 — why a pairwise-MI van Raamsdonk metric cannot select the singlet

The principled metric is van Raamsdonk’s: geometry from entanglement,

$$\ell_{uv}^2 = \left( -\log(I_{uv}/I_{\max}) \right)^2,$$

with  $I_{uv}$  the mutual information of the regions joined by edge  $(u, v)$ . The hope was that seeding the metric from the color-singlet’s entanglement would refine the result toward the pure singlet. It cannot, for a sharp reason.

**Principle 2** (The singlet’s pairwise MI is uniform). *For both candidate three-qutrit singlets — the totally antisymmetric  $\varepsilon$  state  $\sum_{\sigma} \text{sgn}(\sigma) |\sigma(RGB)\rangle$  and the  $\mathbb{Z}_3$  clock state  $|RRR\rangle + \omega|GGG\rangle + \omega^2|BBB\rangle$  — every single-party marginal is maximally mixed and every two-party reduced state has spectrum  $\{0, \frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$ . Hence*

$$S(\rho_X) = \log 3, \quad I(X:Y) = \log 3 \quad \text{for all parties } X, Y,$$

*i.e. the pairwise mutual information is maximal and uniform.*

A metric whose every (non-bulk) edge gets the same  $I = \log 3$  is uniform: it has no structure to distinguish the singlet, and indeed it does not move the result (singlet-overlap 0.737 with the VR metric vs. 0.744 with random jitter). The singlet’s distinguishing content — the relative  $\omega$ -phases — is a genuine *three-party* correlation, invisible to any pairwise marginal. This is the same irreducible-three-body fact that forced the tripartite junction in the first place. Worse, the van Raamsdonk length is manifestly *real* ( $\ell^2 \geq 0$ ), so a VR metric cannot encode a complex phase under any circumstances.

## 5 Choice 4 — the Lorentzian lever (and the photon)

The proton singlet  $[1, \omega, \omega^2]$ ,  $\omega = e^{2\pi i/3}$ , is intrinsically *complex*. A real metric yields real harmonic 1-cochains, whose carried transport is real and cannot represent the  $\omega$ -phases; in our construction those phases enter only through the complex boundary inputs (the  $\mathbb{Z}_3$ -phased neutral pairs already reach  $\approx 0.74$  overlap). To let the *geometry* carry the phases — and lift that ceiling — the metric itself must be complex.

The model already has exactly one source of complex geometry: *Lorentzian* signature. A forward-time worldline edge (endpoints on different layers) carries  $\ell^2 < 0$  (timelike), and the dual Regge action then goes complex,  $\text{Im } S \neq 0$ , so its harmonic spectrum — and the carried transport — becomes complex. The same mechanism realizes the photon hypothesis: an edge driven through  $\ell^2 \rightarrow 0$  is *null* (lightlike), a photon worldline. Thus *the lever that lets the geometry carry the singlet phases is the same lever that emits the photon*.

A first probe (Lorentzian seed, before relaxation) is consistent with this: making the cross-layer worldlines timelike nudges the overlap up ( $0.659 \rightarrow 0.671$ ) and the result toward neutrality ( $|\Sigma_R| : 0.064 \rightarrow 0.027$ ). The full effect requires relaxing the complex (Lorentzian) geometry, where the action’s imaginary part constrains the scale degree of freedom the real part leaves loose — the work now in progress.

## 6 Summary

| Choice                                | Forced because the singlet is three-body                                                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distinct holes (3 independent cycles) | a shared register averages three inputs into the symmetric part; independent cycles + Stokes give $\Sigma_R = -\Sigma_{\text{inputs}}$ (confinement by conservation). |
| Gate the state gradient               | independent cycles carry the inputs <i>exactly</i> , so $r_{\text{state}}$ is floored and its gradient is noise that explodes the step; drop it once converged.       |
| Reject pairwise-MI VR                 | the singlet’s pairwise MI is uniform ( $\log 3$ ); its content is a 3-party phase correlation a real, pairwise metric cannot see or carry.                            |
| Lorentzian (complex) geometry         | the $\omega$ -phases need a complex metric; timelike worldlines make $S$ complex, and the same null-edge limit is the photon.                                         |

The junction is a valid, charge-conserving,  $S_3$ -respecting manifold whose relaxation is fixed and which transports the complex boundary to a result  $\approx 74\%$  aligned with the ideal singlet. The remaining gap is a *phase* gap, and the route to closing it is complex (Lorentzian) geometry, not a finer real metric.